FarmRL: Reinforcement Learning for Crop Input Management and Market Timing

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GitHub: github.com/Harshita772/FarmRL · **Live Dashboard:** farmrl.streamlit.app

1. Problem Context & Target Objective

Wheat cultivation across northwest India follows a 6-month Rabi season (Nov–Apr) divided into $T = 12$ fortnights. Farmers face multi-objective sequential decision trade-offs: applying nitrogen (N), irrigation water (W), and chemical pesticide (P), concluding with post-harvest storage market timing (M). Static government calendar schedules prescribe fixed inputs irrespective of transient weather shocks, causing catastrophic groundwater depletion, aeration hypoxia from waterlogging, fertilizer leaching, and chemical resistance. We map this agricultural decision pipeline into a continuous-state, discrete-action Markov Decision Process (MDP) to maximize farm net income while curbing environmental externalities.

2. Rigorous Reinforcement Learning Formulation

The agricultural environment is formalized as a finite-horizon, discounted Markov Decision Process $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, T \rangle$ with horizon $T = 12$ fortnightly steps:

2.1 State Space ($\mathcal{S}$): Observation Representation

The agent observes a normalized continuous feature vector $s_t \in \mathcal{S} \subset \mathbb{R}^{13}$ (extended to $\mathbb{R}^{14}$ when a rainfall forecast $\hat{R}_t$ is available):

$$s_t = \Big[\tfrac{t}{T},\ m_t,\ \tfrac{N_t^{\text{soil}}}{100},\ p_t,\ \tfrac{R_t^{\text{cum}}}{200},\ \tfrac{N_t^{\text{cum}}}{200},\ \tfrac{W_t^{\text{cum}}}{500},\ I_s^{\text{price}},\ \tfrac{R_{t-1}}{50},\ \tfrac{\text{ET}_{0,t-1}}{100},\ \tfrac{\text{ET}_{a,t}^{\text{cum}}}{\text{ET}_{c,t}^{\text{cum}}},\ \tfrac{c_t^{\text{pump}}}{20},\ e_t^{\text{spray}} \Big]^\top \tag{1}$$

Dimensionality & Components:

- (1) Growth stage $t/T \in [0, 1]$
- (2) Root-zone relative soil moisture $m_t \in [0, 1]$
- (3) Available mineral soil nitrogen $N_t^{\text{soil}}/100$ (kg N/ha)
- (4) Pest infestation density $p_t \in [0, 1]$
- (5–7) Cumulative rain R_t^{cum}, nitrogen N_t^{cum}, irrigation W_t^{cum}
- (8) Agmarknet seasonal grain price index I_s^{price}
- (9–10) Prior rainfall R_{t-1} (mm) & reference ET $\text{ET}_{0,t-1}$ (mm)
- (11) Cumulative water satisfaction ratio $\text{ET}_{a,t}^{\text{cum}}/\text{ET}_{c,t}^{\text{cum}}$
- (12) Marginal tubewell pumping lift cost $c_t^{\text{pump}}/20$ (INR/mm)
- (13) Residual pesticide spray efficacy $e_t^{\text{spray}} \in [0, 1]$
- (14) Forecast precipitation $\hat{R}_t/50$ (when forecast-enabled).

2.2 Action Space ($\mathcal{A}$): Discrete Factored Decisions

Actions are defined over a factored MultiDiscrete($[3, 3, 3, 3]$) space with $|\mathcal{A}| = 3^4 = 81$ joint actions per step:

- *Nitrogen fertilizer:* $N_t \in \{0, 20, 40\}$ kg N/ha
- *Irrigation water:* $W_t \in \{0, 37.5, 75\}$ mm
- *Chemical pesticide:* $P_t \in \{0, 1, 2\}$ doses
- *Post-harvest storage:* $M \in \{0, 1, 2\}$ months (masked until step $t = 11$).

2.3 Reward Function ($\mathcal{R}$): Formulation & Incentives

The reward aligns financial farm profitability with ecological stewardship. Fortnightly reward $r_t = R_t/S_R$ is scaled by $S_R = 10{,}000$ to stabilize policy gradients:

$$R_t = -\Big(c_N N_t + c_W W_t + c_P P_t + C_t^{\text{pump}}(W_t)\Big) - \Big(\lambda_N [N_{t+1}^{\text{soil}} - 80]_+ + \lambda_P P_t\Big) + \mathbb{I}_{[t=11]}\big(\text{Rev}_T - C_{\text{fixed}}\big) \tag{2}$$

Incentive Design:

- (1) Input costs follow CACP: $c_N = 12.9$ INR/kg, $c_W = 2.86$ INR/mm, $c_P = 1{,}183$ INR/dose
- (2) Depth-dependent tubewell pumping penalty $C_t^{\text{pump}}(W_t)$ penalizes aquifer depletion
- (3) Nitrate leaching $\lambda_N = 5$ INR/kg (> 80 kg N/ha) and resistance $\lambda_P = 100$ INR/dose
- (4) Terminal revenue $\text{Rev}_T = Y \cdot P_{\text{sale}}(M) - C_{\text{store}}(M)$ minus fixed $C_{\text{fixed}} = 29{,}450$ INR/ha.

2.4 Environment Dynamics, RL Algorithms, & Baselines

1. Empirically Calibrated Simulation Dynamics: Crop growth, weather, and market dynamics are grounded in authentic Indian agricultural datasets:

- *Weather Replay (NASA POWER):* Daily weather across 39 years (1985–2023) for Ludhiana (Punjab) and Karnal (Haryana). 1985–2016 (32 seasons) serve as training; 2017–2023 (7 unseen seasons $\times$ 30 replicates) are held out for testing.
- *Yield Model (ICRISAT-TCI):* Fitted on 1,719 district-year panel records ($R^2 = 0.512$), coupling a Mitscherlich nitrogen response with FAO-33 water deficit, FAO-66 aeration hypoxia ($\bar{\sigma}^{\text{aer}}$), pest loss ($L_p$), and heat stress:

$$Y = A_{\text{state}} y_{\text{scale}} (1 - e^{-\alpha_N (N_{\text{eff}} + N_0)}) \big(1 - k_y (1 - \tfrac{\text{ET}_a^{\text{cum}}}{\text{ET}_c^{\text{cum}}})\big) \times \prod_{g=1}^{4} (1 - K_{y,g} \bar{\sigma}_g^{\text{aer}}) (1 - L_p D_{\text{pest}})$$

- *Biophysical Dynamics:* Groundwater table depth declines secularly by 0.5 m/yr plus pumping lift drawdown; soil drainage exceeding field capacity triggers root hypoxia and 50% nitrogen leaching; each spray diminishes subsequent efficacy by 25% (pest resistance); stored grain follows Ornstein-Uhlenbeck market price drift.

2. Choice of RL Algorithms: Evaluated for factored action spaces:

- PPO (Proximal Policy Optimization): Primary on-policy actor-critic using clipped surrogate objectives ($\epsilon = 0.2$, GAE $\lambda = 0.95$) and multi-head policy networks for stable convergence under stochastic weather.
- Branching Dueling DQN: Decomposes Q-values into a shared trunk with 4 dueling action branches to eliminate combinatorial action explosion.
- GRPO (Group Relative Policy Optimization): Critic-free policy gradient ($G = 16$ rollouts/state) normalizing advantages within groups to reduce compute overhead.
- CEM (Cross-Entropy Method): Black-box parameter search over policy cutoffs.

3. Baseline Policy Benchmarks: Benchmarked against five domain baselines:

- No-Input Baseline: Natural soil fertility without inputs (17.1 q/ha, 12,234 INR/ha).
- Random Policy: Uniform action selection floor (55,336 INR/ha).
- Govt. Recommended Calendar: Official Punjab extension schedule (74,649 INR/ha).
- Untuned Threshold Rule: Standard ag-extension moisture/pest cutoffs (62,780 INR/ha).
- Tuned Threshold Rule: Grid-searched cutoffs on training seasons (87,232 INR/ha).
- Hindsight Oracle: Upper bound ceiling with clairvoyance of future daily weather and prices (89,654 INR/ha).

Table 1: Held-out test performance across 7 unseen Punjab seasons (2017–2023; mean ± std over 5 seeds, 30 replicates/season).

Policy / Configuration	Profit (INR/ha)	% Orac.	Yield	Irrig.	Sprays
Hindsight Oracle (ceiling)	89,654	100.0%	51.2	225	0.00
Tuned Threshold Rule	87,232	97.3%	50.6	241	0.00
PPO (base v4, 5 seeds)	**86,584 ± 153**	**96.6%**	**50.3**	**220**	**0.00**
GRPO (critic-free, $G = 16$)	**86,223 ± 680**	**96.2%**	**50.1**	**232**	**0.01**
CEM (parameter search)	84,178 ± 192	93.9%	50.4	238	3.46
Branching Dueling DQN	83,617 ± 1,741	93.3%	49.5	221	0.83
Tuned Calendar Rule	81,072	90.4%	48.4	300	0.00
Recommended Calendar (Govt)	74,649	83.3%	45.8	300	1.00
Untuned Threshold Rule	62,780	70.0%	45.9	236	11.80
Random Policy	55,336	61.7%	44.9	458	11.80
No-Input Baseline	12,234	13.6%	17.1	0	0.00

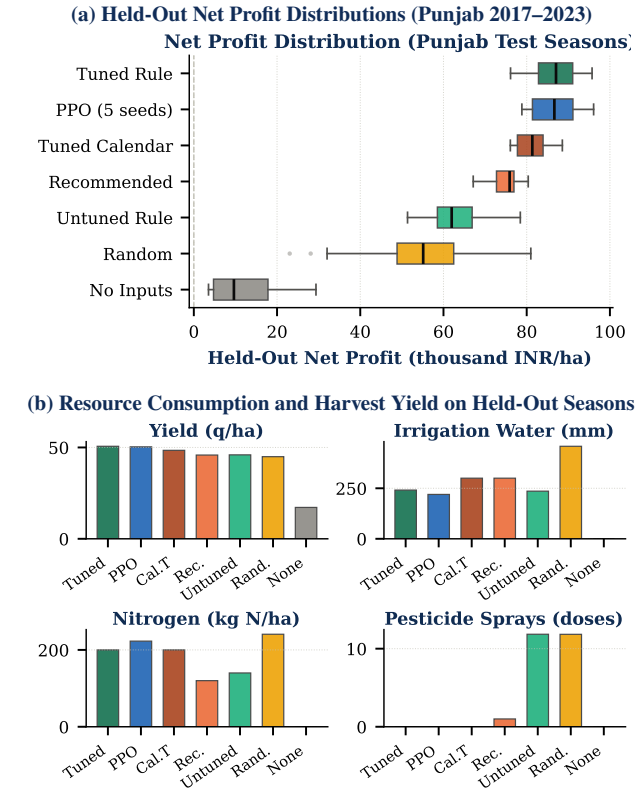


Figure 1: Held-out policy benchmarks across 7 unseen Punjab seasons (2017–2023): (a) PPO achieves 96.6% of the hindsight oracle ceiling (+11,935 INR/ha over calendar advice). **(b)** Input efficiency: PPO matches peak yield (50.3 q/ha) while saving 26.7% irrigation water (220 vs 300 mm) with zero routine sprays.

3. Empirical Evaluation, Ablations, & Best Model

Table 1 and Figure 1 report held-out benchmarking. Below we detail the 12 systematic ablation studies across 5 seeds:

1. Baseline Gap and Shapley Attribution: Base PPO (2×64 MLP, GAE $\lambda = 0.95$) achieves 86,584 ± 153 INR/ha, outperforming Punjab extension advice by +11,935 INR/ha (+16.0%) and cutting irrigation water consumption by 26.7% (220 vs 300 mm). Shapley decomposition across all 24 action coalitions reveals that 62% of this gain originates from adaptive irrigation timing and 31% from state-dependent nitrogen split timing. PPO trails the static tuned rule by −648 INR/ha due to rainfall uncertainty.

2. Optimization Invariance: Sweeping learning rate schedules, observation normalization, orthogonal initialization, reward discount $\gamma = 1.0$, and training up to 1M steps shifted profits by $< 0.1\%$, proving that the gap was informational rather than algorithmic. Branching DQN (83,617) and CEM (84,178) lagged behind PPO.

3. Action Space Discretization: Upgrading to a 4-level irrigation grid $\{0, 25, 50, 75\}$ mm paired with a joint $N \times W$ action head boosted PPO to 86,821 ± 315 INR/ha, outperforming its own retuned rule across all 5 seeds.

4. Pest Severity Response & Selective Spraying: Under realistic pest damage ($L_p \geq 0.30$), PPO decisively outperforms the retuned rule across all seeds by +2,601 to +3,827 INR/ha (Figure 2a). PPO executes resistance-aware selective spraying (3.59 vs 6.32 doses at

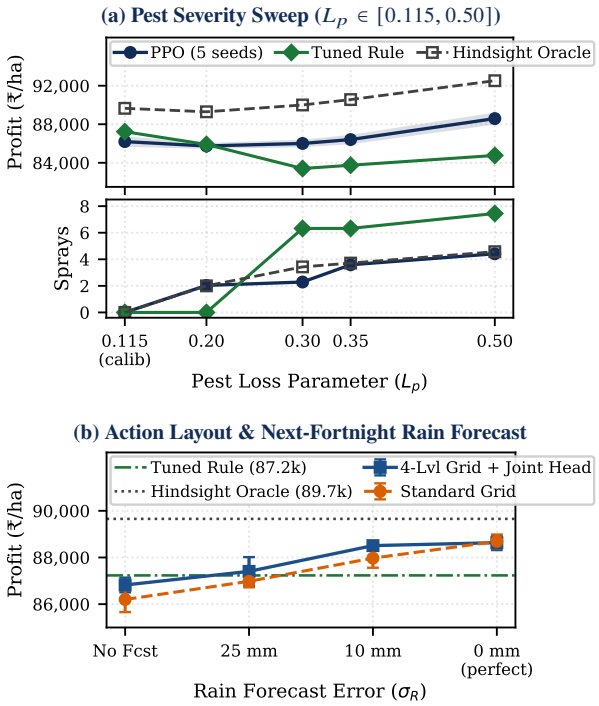


Figure 2: Performance sweeps with zero-overlap legends: (a) For pest loss $L_p \geq 0.30$, PPO beats the retuned rule across all seeds (+2.6k to +3.8k INR/ha) via selective spraying. **(b)** Combining a 4-level water grid, joint head, and a 10 mm rain forecast achieves 88,508 ± 81 INR/ha (98.9% of oracle).

$L_p = 0.35$), preventing pesticide resistance collapse.

5. Best Model Discovery (Weather Foresight): Oracle decomposition proves future weather clairvoyance is worth +5,367 INR/ha vs +188 INR/ha for price foresight. Providing a 14-day rainfall forecast with error $\sigma_R = 10$ mm to the 4-level joint agent produced our **best overall model:** 88,508 ± 81 **INR/ha (98.9% of oracle ceiling)**, outperforming the tuned rule by +1,963 INR/ha across all 5 seeds (Figure 2b).

6. Critic-Free GRPO and Cross-State Generalization: Critic-free GRPO matched PPO (86,223 vs 86,584 INR/ha) while reducing training runtime by 6%. Zero-shot transfer from Punjab to Karnal (Haryana) achieved 74,303 ± 463 INR/ha (96.2% of Haryana oracle), exceeding Haryana's local tuned rule (73,831 INR/ha) without retraining.

Table 2: Systematic ablation and experiment breakdown across 5 seeds: progression from base PPO to the best foresight model.

Study / Ablation	Intervention / Mechanism	Profit (INR/ha)	Gap to Rule	% Orac.	Win
01: Base PPO	3-lvl grid, separate heads	86,584 ± 153	−648	96.6%	0/5
03: Action Layout	4-lvl water + joint $N \times W$	86,821 ± 315	+621	96.9%	5/5
05: Severe Pest Loss	$L_p = 0.35$ (selective spray)	86,410 ± 492	+2,664	95.4%	5/5
07: Weather Forecast	Standard grid, $\sigma_R = 10$ mm	87,972 ± 412	+586	98.3%	4/5
12: Best Model	**4-lvl grid + $\sigma_R = 10$ mm fcst**	**88,508 ± 81**	**+1,963**	**98.9%**	**5/5**
12: Perfect Weather	4-lvl grid + clairvoyant rain	88,635 ± 317	+1,403	99.1%	5/5
Transfer: Haryana	Zero-shot to Karnal climate	74,303 ± 463	+472	96.2%	5/5

4. Conclusion

FarmRL demonstrates that reinforcement learning discovers water-conserving, resistance-mitigating input policies in complex agricultural domains. While tuned heuristics are competitive in static scenarios, RL policies decisively excel when pest pressures are non-trivial and when conditioning on short-term weather forecasts.

References

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